EDUC 697A Literature Synthesis

Topic: Universal Design for Learning (UDL) and K–12 English Language Learners

1. Research Question

How can Universal Design for Learning (UDL) support the academic learning and classroom participation of K–12 English language learners across different subject areas?(Sub-Question)

How do UDL strategies support the academic learning of K–12 English language learners?

How does UDL influence the classroom participation and engagement of K–12 English language learners?

2.Subset of 10 Focal Studies

Recommended 10 Focal Studies

1. Tai, K. W. H. (2022). Translanguaging as inclusive pedagogical practices in English-medium instruction science and mathematics classrooms for linguistically and culturally diverse students. *Research in Science Education, 52*(3), 975–1012. https://doi.org/10.1007/s11165-021-10018-6

Author : Tai, K. W. H.

year : 2022

Theme :Translanguaging as inclusive pedagogical practices in English-medium instruction science and mathematics classrooms for linguistically and culturally diverse students. Research in Science Education, 52, 975–1012.

URL:https://link.springer.com/article/10.1007/s11165-021-10018-6

Kevin Tai from university college of London examines English-medium science and mathematics classrooms for ESL students. By analyzing how teachers use multilingual and multimodal resources to help students understand academic content

and participate in classroom interaction. This study is highly relevant to my research question because it demonstrates how educators can support both academic learning and classroom participation through approaches that closely align with UDL principles, especially multiple means of representation and engagement. Although this is not directly a UDL intervention study, the teaching practices share many similarities with the UDL framework. This study provides an example of what an inclusive and flexible classroom can look like, and it helps build a transition to the idea that UDL may be particularly well suited to supporting K–12 English language learners across different subject areas.

2.

Author: Ok, M. W., Rao, K., Bryant, B. R., & McDougall, D.

Year: (2017)

Theme: Universal Design for Learning in Pre-K to grade 12 classrooms: A systematic review of research. *Exceptionality, 25*(2), 116–138

URL: https://doi.org/10.1080/09362835.2016.1196450

Systematic review of K–12 UDL research. Useful for the overall evidence base and for identifying mixed effects and reporting limitations. **Ok et al. (2017) – K–12 UDL Systematic Review**

This review examines UDL research conducted in K–12 settings and reports potential benefits related to engagement, as well as helping studetns with access to the general curriculum, academic outcomes, and social outcomes through UDL. It is important to my study because it provides broad evidence that UDL can support diverse learners in K–12 classrooms. However, the review also reports inconsistencies in how UDL was implemented and measured across studies.(now strating to introduct how UDL was practived in general classrooms)

3.

Author: King-Sears, M. E., Stefanidis, A., Evmenova, A. S., Rao, K., Mergen, R. L., Owen, L. S., & Strimel, M. M.

Year:(2023)

Theme: Achievement of learners receiving UDL instruction: A meta-analysis. *Teaching and Teacher Education, 122*,

URL: 103956. https://doi.org/10.1016/j.tate.2022.103956

Margaret E. King-Sears and her team interpreted this Meta-analysis reporting a moderate positive combined effect on academic achievement under UDL-based learning method. This meta-analysis examines the academic achievement of learners who received UDL-based instruction. The combined results suggest a moderate positive effect of UDL on academic achievement. This study mainly contributes to my first sub-question because it provides **quantitative** evidence that UDL-based instruction may improve student learning outcomes. Limits: UDL for all students, not particularly for ESL students.

4.

Author: Katz, J. N.

Year: (2013)

Theme: The Three Block Model of Universal Design for Learning (UDL): Engaging students in inclusive education. *Canadian Journal of Education/Revue canadienne de l'éducation, 36*(1), 153–194.

URL:https://www.researchgate.net/publication/286055447_The_Three-Block_model_of_universal_design_for_learning_UDL_Engaging_students_in_inclusive_education

Large Grades 1–12 study reporting increased active engagement, peer interaction, autonomy, and inclusion.

This study examines the Three Block Model of UDL in Grades 1–12 and it strongly focuses on student engagement and inclusion. The findings indicate increases in active engagement, peer interaction, autonomy, and inclusive classroom participation. This article is particularly useful for my second sub-question about participation and engagement. Although it is not specifically focused on English language learners, it provides evidence that UDL can create more participatory and inclusive learning environments.

5.

Author: Bray, A., Devitt, A., Banks, J., Sanchez Fuentes, S., Sandoval, M., Riviou, K., Byrne, D., Flood, M., Reale, J., & Terrenzio, S

Year: (2024)

Theme: What next for Universal Design for Learning? Asystematic Literatire review of techonology in UDL implementations at second level

URL:

https://bera-journals.onlinelibrary.wiley.com/doi/10.1111/bjet.13328

Focal study explanation:

Bray et al. (2024) conducted a systematic literature review of 15 empirical studies examining how digital technology has been used to support UDL in secondary education. The review found that most UDL implementations focused on **multiple means of representation**, particularly by giving students different ways to access learning materials through multimedia and technology. However, the authors found much less research on using technology to support **engagement, action and expression, self-regulation, communication, and collaboration**. Bera Journals This study is relevant to my research question because it shows how UDL can create more flexible learning opportunities across secondary-school subjects and also identifies areas where classroom participation could be strengthened. Although the study is not specifically about English language learners, its findings are useful for considering

how technology-based UDL practices could reduce barriers for ELL students who may benefit from multiple ways of accessing content and expressing their learning.

6.

Author: Williams, M., Tang, K.-S., & Won, M.

Year:(2019).

Theme:ELL's science meaning making in multimodal inquiry: A case-study in a Hong Kong bilingual school. *Asia-Pacific Science Education, 5*(1), Article 3

URL: https://doi.org/10.1186/s41029-019-0031-1

ELL's science meaning making in multimodal inquiry: a case-study in a Hong Kong bilingual school , Published: 12 February 2019 Melanie Williams, Kok-Sing Tang &Mihye Won

https://link.springer.com/article/10.1186/s41029-019-0031-1

Scientifical support of learning method for ESL students in HongKong, in terms of **Systemic functional linguistics influences social semiotics , The function of language , Language-based theory of learning**

Williams, Tang, and Won (2019) – ELL Science Meaning-Making

This study investigates Grade 5 English language learners in a bilingual science classroom in Hong Kong. It examines how students use multiple modes such as speech, writing, drawing, gestures, and models to construct scientific meaning. The findings show that multimodal learning can help students understand academic content and express their knowledge even when their English proficiency is still developing. This study is highly relevant to UDL because it supports the idea of providing multiple means of representation and multiple ways for students to demonstrate understanding.(Other learning method that UDL could include, more specifically, by social semiotics in the school system)

7.

Author: Meskill, C., Nilsen, J., & Oliveira, A

Year: (2019)

Theme: Intersections of language, content, and multimodalities: Instructional conversations in Mrs. B's sheltered English biology classroom. *AERA Open, 5*(2), 1–15

URL: https://doi.org/10.1177/2332858419850488

Intersections of Language, Content, and Multimodalities: Instructional Conversations in Mrs. B's Sheltered English Biology Classroom Carla Meskill, Jennifer Nilsen, and Alan OliveiraView all authors and affiliations, First published online May 17, 2019

https://journals.sagepub.com/doi/10.1177/2332858419850488

Multimodal Mediation and ELs

For the growing number of ELs in U.S. schools, teacher mediation of content through multiple perceptual modalities has been well established as supportive to the development of language/content

Real classroom learning curriculum / conversations

Meskill, Nilsen, and Oliveira (2019) – Sheltered English Biology Classroom
This study examines instructional conversations in a high school sheltered biology classroom for English learners. The teacher used visuals, gestures, digital tools, discussion, and multiple forms of explanation to support both language development and biology learning. The study shows that English learners can access complex academic content when teachers use multimodal and flexible instructional strategies. These practices align closely with UDL, especially multiple means of representation.

8.

Author: Michalovich, A.

Year: (2023).

Theme: The social drama of digital multimodal composing: A case study with emergent bi/multilingual newcomer students. *Journal of Adolescent & Adult Literacy, 66*(6), 334–343.

URL: https://doi.org/10.1002/jaal.1286

Michalovich (2023) – Digital Multimodal Composing

This study focuses on emergent multilingual newcomer students who participated in digital multimodal composing activities. Students were able to utlize different modes including language, images, digital media, and other forms of expression through UDL, to communicate their ideas. The findings highlight student engagement, identity, social connection, and investment in classroom learning. This study is relevant to my research because it supports the UDL principle of multiple means of action and expression and shows how flexible forms of communication may increase participation.

9.

Author: Li, G

Year:. (2018).

Theme: From stigma to strength: A case of ESL program transformation in a Greater Vancouver high school. *BC TEAL Journal, 3*(1), 63–76.

From Stigma to Strength: A Case of ESL Program Transformation in a Greater Vancouver High School

Guofang LiUniversity of British Columbia, Published 2018-10-12

https://ojs-o.library.ubc.ca/index.php/BCTJ/article/view/303/348

Li (2018) UBC – From Stigma to Strength

This article examines the transformation of the ESL program in a Greater Vancouver high school. It discusses how traditional pull-out ESL programming could create separation and stigma for English language learners, while a more inclusive and

culturally responsive program better supported students' academic, linguistic, and socio-emotional needs. This study is for understanding the importance of inclusion, belonging, and program design for English language learners. It is not a direct UDL study, it also supports the broader UDL idea of redesigning the learning environment so that students can participate more engagingly in the regular classroom community.

1**0.**

Author: Panhwar, F. Y., Memon, I., & Baluch, N.

Year: (2018).

Theme: Challenges faced by learners in ESL: A case study of Institute of English Language and Literature, University of Sindh, Jamshoro, Pakistan. *The Sindh University Journal of Education, 47*(2), 1–24

Panhwar et al. (2018) – Challenges Faced by ESL Learners

This case study explores challenges experienced by ESL learners, including limited confidence, language difficulties, motivation, teaching practices, and curriculum-related concerns. It is useful for identifying barriers that can interfere with student learning and participation. These barriers help explain why flexible and supportive teaching approaches such as UDL may be valuable for English language learners. However, this study is based in higher education rather than K–12, so it should be used mainly as background evidence rather than as one of the strongest studies answering my research question.

3.

Theme Synthesis Table

Theme (emerging claim)	**Studies and shared evidence**	**Difference or limitation**	**What this suggests for the research question**
Theme 1	**Tai (2022), Williams et al. (2019), and Meskill et al. (2019)** show that using multiple languages through visuals, gestures, models, discussion, and other modes can help English learners understand complex academic content in science and mathematics. **Bray et al. (2024)** also found that multiple means of representation were commonly used in technology-supported UDL at the secondary level.	Tai, Williams, and Meskill examine multilingual or English learner classrooms, but they are not all direct UDL intervention studies. Bray et al. focuses on UDL more broadly and is not specifically about ELLs.	UDL may support ELL academic learning by reducing language-related barriers and providing multiple ways to access and understand subject content across different courses.
Theme 2	**Katz (2013)** reported increased engagement, peer interaction, autonomy, and inclusion in UDL classrooms. **Michalovich (2023)** showed that multimodal	Katz is not specifically focused on ELLs, while Michalovich and Li are not direct UDL	UDL may increase classroom participation by giving English

	composing supported newcomer students' participation, identity, and investment in learning. **Li (2018)** also showed that more inclusive and culturally responsive ESL programming can strengthen belonging and participation.	studies. The evidence for participation is therefore strong conceptually, but not always based on UDL-specific ELL interventions.These multi modal practices align with UDL principles.	learners more choice in how they communicate, interact, and demonstrate learning rather than relying only on traditional English-language responses.
Theme 3	**Ok et al. (2017)** found generally promising K–12 outcomes for engagement, curriculum access, and learning, while **King-Sears et al. (2023)** reported an overall positive effect of UDL on academic achievement. **Bray et al. (2024)** found that UDL	Studies use different definitions, technologies, populations, and measures of UDL. Some focus on students with disabilities rather	UDL appears promising for supporting ELL learning and participation, but stronger ELL-specific K–12 research is

	implementation often emphasized representation more than engagement or action/expression.	than ELLs, and there is less direct evidence specifically examining UDL with K–12 English learners across subjects.	needed to determine which UDL strategies are most effective and how consistently they should be implemented across subject

4.**AI usage statement**

AI was used to help locate and organize a few potential references, format citations, and grammar support for this assignment. I independently reviewed the original sources and verified all citations, links, study findings, and thematic claims before including them in my work. Some references that were initially found through ChatGPT contained a ChatGPT-related tracking tag at the end of the URL, such as "source=chatgpt.com"; these tags only indicate how the link was accessed and are not part of the original academic source.

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